

JULY 21 2005

Acoustic Nonlinearity of an Orifice

Uno Ingard; Hartmut Ising



J. Acoust. Soc. Am. 42, 6–17 (1967)

<https://doi.org/10.1121/1.1910576>



CrossMark

Articles You May Be Interested In

Total pressure in thermionic orificed hollow cathodes: Controlling mechanisms and their relative importance

J. Appl. Phys. (January 2022)

Two-dimensional model of orificed micro-hollow cathode discharge for space application

Phys. Plasmas (August 2013)

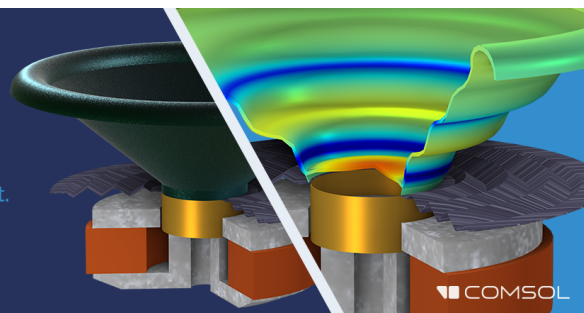
Physical understanding of gas-liquid annular flow and its transition to dispersed droplets

Physics of Fluids (July 2016)

Take the Lead in Acoustics

The ability to account for coupled physics phenomena lets you predict, optimize, and virtually test a design under real-world conditions – even before a first prototype is built.

» Learn more about COMSOL Multiphysics®



Acoustic Nonlinearity of an Orifice

UNO INGARD

*Research Laboratory of Electronics and Department of Physics, Massachusetts Institute of Technology,
Cambridge, Massachusetts 02139*

HARTMUT ISING

Institut für Technische Akustik, Technische Universität, Berlin, Germany

The acoustic nonlinearity of an orifice in a plate has been investigated by measuring simultaneously the oscillatory flow velocity in the orifice and the acoustic-pressure fluctuations producing the flow. The relation between the pressure and velocity amplitudes, which is linear at sufficiently low pressures, is found to approach a square-law relation at large velocity amplitudes. By evaluating the phase relationship between the fundamental harmonic components of pressure and velocity, the acoustic-orifice impedance is determined. In the square-law region of the pressure/velocity relation, the resistive component of the orifice impedance dominates and is proportional to the velocity amplitude. This method of measurement, which is particularly useful at high amplitudes, is supplemented by measurements of sound transmission and frequency response of the orifice plate so that an impedance curve can be constructed over an extended range of amplitudes. An analogous program of measurements has been carried out to study the influence of a superimposed steady flow on the acoustic-orifice impedance. Finally, as an application of the results obtained, the absorption characteristics of resonator absorbers for high-intensity sound are discussed.

INTRODUCTION

AFTER Sivian¹ reported, in 1935, that the acoustic resistance of an orifice in a plate depends on the strength of the incident sound wave and after a similar nonlinear behavior of the orifice reactance had been observed,² several studies were made to determine the amplitude dependence of the nonlinearity and the dependence of the orifice geometry. For example, on the basis of an extensive study at sound-pressure levels (SPL's) up to 130 dB and velocity amplitudes in the orifice up to 10 m/sec, a stronger than linear dependence of the nonlinear resistance on the velocity amplitude in the orifice was reported.³ Measurements at higher velocities^{4,5} have led to a substantially linear dependence of the orifice resistance on the velocity amplitude, and a similar result has been obtained also for oscillatory

orifice flow of liquids.^{6,7} Recently, other results^{8,9} have been reported in measurements on perforated plates at SPL's up to 170 dB.

Apart from the experiments with liquids, these previous studies have all been based on the standing-wave method of impedance measurement. In view of the apparent differences in the reported results, we wish to reexamine here the problem of the acoustic-orifice nonlinearity and describe a more direct method of measurement, one that involves the simultaneous measurement of the oscillatory-flow velocity in the orifice and the acoustic-pressure fluctuations producing this flow. This method not only is comparatively simple but gives a better insight into the problem than does the standing-wave method of impedance measurement.

The influence of a superimposed steady flow on the orifice impedance has been studied previously by

¹ L. J. Sivian, *J. Acoust. Soc. Am.* **7**, 94–101 (1935).

² R. H. Bolt, S. Labate, and U. Ingard, *J. Acoust. Soc. Am.* **21**, 94–97 (1949).

³ U. Ingard, *J. Acoust. Soc. Am.* **25**, 1037–1061 (1953).

⁴ D. A. Bies and O. B. Wilson, Jr., *J. Acoust. Soc. Am.* **29**, 711–714 (1957).

⁵ F. Barthel, *Frequenz* **12**, 1–11 (1958).

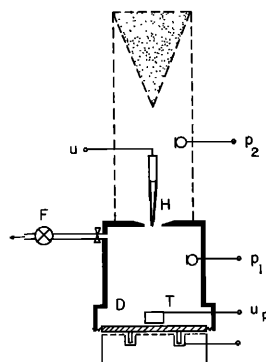
⁶ G. B. Thurston, L. E. Hargrove, Jr., and B. D. Cook, *J. Acoust. Soc. Am.* **29**, 992–1001 (1957).

⁷ G. B. Thurston, *J. Acoust. Soc. Am.* **30**, 452–455 (1958).

⁸ A. W. Blackman, *ARS J.* **30**, 1022–1028 (1960).

⁹ P. A. Marino, Jr., N. Bohn, and G. D. Garrison (to be published).

FIG. 1. Experimental arrangement. D: Piston driver. T: Piston-velocity pickup. H: Hot wire. F: Fan. The external tube (shown dashed) was removed in some of the experiments.



several investigators.^{10,11,5} The experimental data on acoustic resistance are in qualitative agreement with the acoustic resistance derived from the known static pressure/flow characteristic of the orifice by considering sound as a quasistatic modulation of the static flow.^{10,5} Our experimental arrangement lends itself readily to the study of this problem, and we have included it in our reexamination of the orifice problem. Observations of the oscillatory-flow velocity in the orifice have been made as a function of the velocity of a steady superimposed flow, and we include a case not previously considered in which the oscillatory-flow amplitude is of the same order of magnitude as the steady flow. The similarity between the large-amplitude acoustic response of an orifice and the influence of the superimposed flow is discussed.

I. EXPERIMENTAL ARRANGEMENT

The experimental arrangement used in our experiments is indicated in Fig. 1. The orifice plate is mounted on one end of a circular-tube section that has an inside diameter of 95 mm and a length of 70 mm. The orifice diameter is 7 mm, and the thickness of the plate at the orifice is tapered from 6 mm down to 0.1 mm. An acoustic driver unit is attached to the other end of the tube. The driver, a specially designed electromagnetically driven square piston with 100-mm sides, is attached by means of a rubber support to its housing. The driver is provided with a separate dynamic pickup to measure the velocity of the piston. The cavity thus formed between the driver and the orifice plate has a volume of 710 cm³. (In a modified version of the apparatus, the volume was 640 cm³.)

In some of our experiments, a second tube section was attached to the outside of the orifice plate to "collect" the sound that was transmitted through the orifice into a plane wave that was absorbed at the end of the tube by means of a porous wedge absorber. The inside diameter of this second tube section was 66 mm. (In one set of experiments, a tube diameter of 62 mm was used.)

¹⁰ P. J. Westervelt and P. W. Sieck, *J. Acoust. Soc. Am.* **22**, 680(A) (1950).

¹¹ U. Ingard, *J. Acoust. Soc. Am.* **31**, 1202-1212 (1959).

To enable measurements in the presence of a superimposed steady flow, a small outlet in one of the cavity walls can be connected to a suction device with continuous-speed regulation.

The SPL is measured by means of a small condenser microphone inserted in the cavity wall, and the flow velocity in the orifice is determined with a thin hot-wire probe placed at the center of the orifice. The hot-wire instrument, which is of the constant-temperature variety, is provided with a linearization device so that the output from the instrument will be proportional to the magnitude of the velocity in the range 1-50 m/sec to an accuracy of $\pm 3\%$. The sound-pressure and velocity signals can be observed simultaneously on a double-beam oscilloscope and can also be recorded on a logarithmic recorder. From an analysis of the photographic records of the oscilloscope traces, the relationship between the fundamental-harmonic components of pressure and velocity can be determined. The oscillator driving the piston source can be regulated by means of appropriate feedback circuitry to keep the amplitudes of either the cavity SPL or the orifice velocity constant. This regulation was used in response measurements in which the frequency of the driver was continuously varied.

II. HOT-WIRE MEASUREMENTS OF OSCILLATORY FLOW IN THE ORIFICE

In the first set of experiments, the external tube, indicated by dashed lines in Fig. 1, was not attached to the plate so that the sound from the orifice was emitted into free space. The piston was driven at a frequency of 150 Hz, and the SPL's in the cavity in front of the aperture plate were varied between 110 and 162 dB, corresponding to sound-pressure amplitudes 88.5-3500 dyn/cm². The SPL in the cavity and the velocity of the oscillatory-orifice flow in the orifice were measured as described above. Typical oscilloscope traces of the pressure and velocity at some different SPL's are given in Fig. 2. Note that the pressure and velocity scales at the different levels are not the same; the actual measured peak values are given in Table I and the peak value of the velocity is shown in Fig. 3 as a function of the pressure amplitude.

The driving sound pressure is essentially sinusoidal at all levels that are considered. At low levels, repre-

FIG. 2. Typical oscilloscope traces of the sound pressure p in the cavity (see Fig. 1) and the corresponding velocity u of the flow in the orifice at some different SPL's in the cavity. Notice the distortion of the velocity curves with increasing SPL and the gradual change in the phase difference between pressure and velocity. (The velocity curves, obtained from the hot-wire instrument, are proportional to the *magnitude* of the velocity, so they appear rectified in the Figure.)

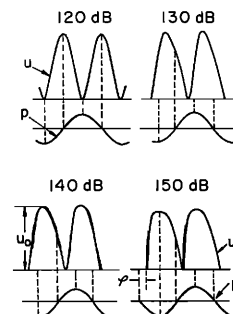


TABLE I. Measured values of the amplitudes of the "driving" sound pressure and the corresponding velocity amplitude in the orifice.

$(\text{dyn/cm}^2)^2$	p_1 (dB)	u_0 (m/sec)	$p/(\rho u_0^2/2)$
280	120	2.70	
500	125	4.85	
900	130	7.25	
1 600	135	10.8	
2 800	140	15.0	1.97
5 000	145	20.5	1.89
9 000	150	27.0	1.96
12 600	153	31.5	2.0
20 000	157	41.0	1.89

sented by the 120-dB curve in Fig. 2, the velocity trace is sinusoidal and indicates linear response. At higher levels, however, the velocity traces are markedly distorted. This distortion is related to the nonlinear (quadratic) relationship between the amplitudes of pressure and velocity at higher levels shown in Fig. 3. The transition from linear to quadratic response occurs at a velocity amplitude u_t in the orifice, which in our case is ~ 10 m/sec. At velocities greater than this value, we see from Table I that the relation between the SPL p in the cavity (see Fig. 1) and the velocity amplitude u_0 in the orifice is approximately given by

$$p \sim \rho u_0^2, \quad u_0 > u_t. \quad (1)$$

This result, as we see later, can be interpreted on the basis of Bernoulli's law for time-dependent flow.

In the linear régime, the flow about the orifice is essentially irrotational and the acoustic driving pressure is balanced predominantly by the acceleration of the flow in the vicinity of the orifice. At the frequencies considered here, the influence of viscous drag on the sound-pressure/velocity relation is small. This is apparent from Fig. 2, which shows that at low levels the pressure and velocity curves are practically 90° out of phase.

In the nonlinear régime, flow separation takes place and the flow takes the form of an oscillatory jet through the orifice. Thus, at every moment, the flow pattern is asymmetrical on the two sides of the orifice plate. On one side, we have an essentially irrotational in-flow, whereas on the other side a jet is formed. After half a cycle, these flow patterns are interchanged. This

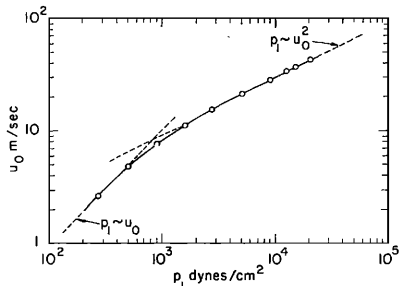


FIG. 3. Peak velocity in the orifice as a function of pressure amplitude in the cavity. Frequency: 150 Hz. At low amplitudes, the relationship is linear; at higher amplitudes, it is quadratic.

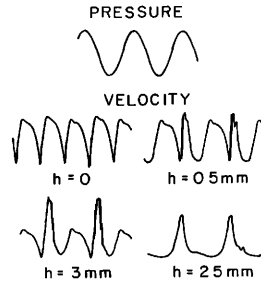


FIG. 4. Hot-wire records of the axial velocity of the oscillatory flow at some different axial distances away from the orifice. SPL: 140 dB. The asymmetry of the velocity traces indicates flow separation and jet formation.

asymmetry of the flow is easily demonstrated by placing the hot wire outside the plane of the orifice. The corresponding velocity trace, shown in Fig. 4 at some different axial positions away from the orifice, is no longer symmetric. In fact, the in-flow velocity at distances larger than 25 mm from the orifice, in this particular case, is too small to be detected by the hot wire, and only the half-cycle during which the jet flow reaches the hot wire is recorded, as can be seen in Fig. 4.

We describe the acoustic nonlinearity in terms of an orifice impedance. The concept of impedance refers to the relationship between sound pressure and velocity at a particular frequency. Thus, if we wish to introduce an orifice impedance that refers to the driving frequency $(\omega/2\pi)$ of the sound pressure in the cavity, we must first determine the fundamental harmonic-frequency component of the velocity in the orifice. The distortion of this velocity, as shown in Fig. 2, is characterized by a marked difference in slope of the increasing and decreasing portions of the curve. The essential feature of this waveform can be described by the trapezoidal curve form shown in Fig. 5. The parameters defining this curve are the positions x_1 and x_2 of the corners of the curve and the amplitude y , which is so chosen that the area enclosed by the trapezoidal curve is the same as that of the actual curve. This amplitude turns out to be practically the same as the peak amplitude u_0 referred to in Fig. 2.

Fourier decomposition of the characteristic curve in Fig. 5 leads to the following expressions for the amplitude u_1 and phase angle ϕ (see Fig. 5) of the fundamental-harmonic component of the velocity

$$u_1 = (2y/\pi)(A^2 + B^2)^{1/2}, \quad \tan \phi = B/A; \quad (2)$$

$$A = \frac{\sin x_1}{x_1} + \frac{\sin x_2}{\pi - x_2}, \quad B = \frac{1 + \cos x_2}{\pi - x_2} - \frac{1 - \cos x_1}{x_1}.$$

In accordance with this procedure of analyzing the velocity curve, the results obtained from the experimental velocity traces shown in Fig. 2 are given in Table II.

We see that the amplitude u_1 of the fundamental-harmonic component is somewhat smaller than the peak value u_0 of velocity. More important, however, is the fact that the fundamental-harmonic component leads the actual velocity curve by a certain phase angle ϕ .

TABLE II. Measured values of the parameters defining the waveform in Fig. 5 of the velocity in the orifice.

SPL (dB)	x_1	x_2	Velocity amplitude of fundamental (u_1/u_0)	Phase ϕ (see Fig. 5)
120	0.5π	0.5π	1	0
135	0.27π	0.38π	0.90	12
140	0.22π	0.36π	0.91	16.5
150	0.17π	0.32π	0.91	19
157	0.14π	0.28π	0.90	20

Accounting for this phase lead, we can now determine from the experimental data (see Fig. 2) the phase difference ϕ_1 between the fundamental-harmonic components of the pressure and velocity. The results are given in Fig. 6, where the phase difference ϕ_1 is plotted as a function of the SPL in the cavity. In the same Figure is shown the "phase difference" φ (see Fig. 2) between the sound-pressure and the total-velocity curve.

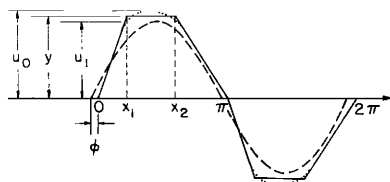
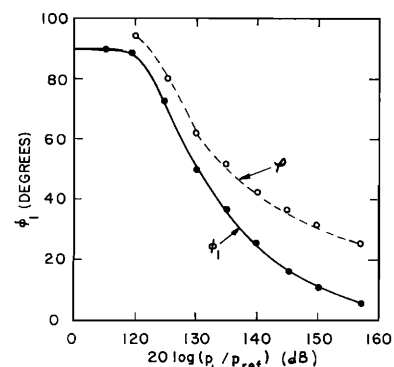


FIG. 5. Shape of the actual velocity curve (. . .) is approximated by a trapezoidal curve (—) from which the fundamental-harmonic component (— —) is determined.

Having obtained the amplitude and phase relationship between the fundamental-harmonic components of the pressure and velocity, we can now determine the orifice impedance Z . The ratio between sound pressure in the cavity and velocity in the orifice is given by the sum of Z and the radiation resistance corresponding to the sound field generated by the velocity in the orifice. Actually, the reactive part of the radiation impedance is included in the orifice impedance, since it represents the effect of the inertial mass of the local air motion in the vicinity of the orifice. It is usually thought of as a mass "end correction." A similar correction applies also to the inside of the orifice plate. The radiation resistance of the orifice, which is approximately the same as that for a plane piston in a rigid baffle—namely, $\frac{1}{16} (kd)^2$ —is much smaller even than the viscous resistance of the orifice at the frequencies considered here and can be neglected. Consequently, the orifice impedance can be determined simply from the measurement of the driving sound pressure in the cavity and the velocity in the orifice. Note, however, that, in order to include the inside mass end correction of the orifice, the driving sound pressure must be measured at a point that is outside of the near flow field of the orifice, which extends two or three orifice diameters from the center of the orifice.

FIG. 6. Phase angle ϕ_1 by which the fundamental-harmonic component of the velocity lags behind sound pressure. The total velocity curve (see Fig. 2) lags behind sound pressure by the phase angle φ , also shown here as a function of the SPL of the driving sound field.



Thus, from the measurements of the fundamental components of the pressure and the velocity and the phase angle between them, p_1 , u_1 , and ϕ_1 , the magnitude of the orifice impedance $|Z|$ and the resistive and reactive components R and X —

$$|Z| = p_1/u_1, \quad R = |Z| \cos \phi_1, \quad X = |Z| \sin \phi_1, \quad (3)$$

—can be determined and they are plotted in Fig. 7 as a function of the peak value u_0 of the velocity in the orifice. (The corresponding sound-pressure amplitude is obtained from Fig. 2.) Perhaps the most interesting aspect of these curves is the behavior at large values of the velocity amplitude in the orifice, larger than, say, 10 m/sec. In this region, the magnitudes of the impedance and the resistance are about equal and proportional to the velocity amplitude in the orifice. In fact, the $|Z|$ and R curves almost coincide with the line $R = \rho u_0$, also shown in the Figure. We return to this below.

With the particular hot-wire instrumentation used in the experiment, the accuracy of the velocity measurements is not particularly good at low velocities, and the impedance data in Fig. 7 are limited to comparatively large velocity amplitudes. Although the accuracy in the measurement of the absolute magnitude $|Z|$ of the impedance is adequate, particularly at large amplitudes, our determination of the phase difference ϕ_1 between the fundamental components of pressure and velocity cannot be considered accurate enough to make the accuracy

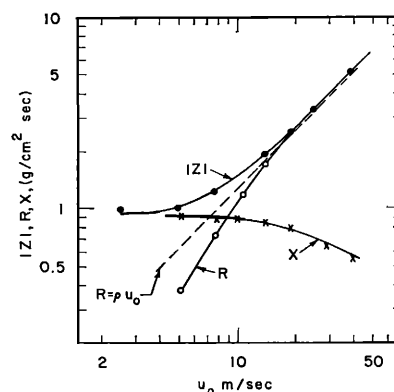


FIG. 7. Magnitude $|Z|$ and the resistive and reactive components R and X of the orifice impedance as a function of the velocity amplitude u_0 in the orifice. For comparison, the line $R = \rho u_0$ is also shown.

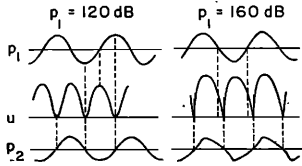


FIG. 8. Typical oscilloscope traces of the driving sound pressure p_1 , orifice velocity u , and transmitted plane-wave pressure p_2 , illustrating linear and nonlinear behavior.

in R and X as good as that of $|Z|$, particularly at angles close to 90° (for R) and 0° (for X). We supplement these impedance data by other results obtained from the measurements discussed below. This will lead to impedance curves that cover a wider range of velocity amplitudes than that in Fig. 7.

III. TRANSMISSION THROUGH ORIFICE

In the experiment described here, a tube with a diameter of 62 mm was added to the orifice plate and provided with a " ρc termination," as shown in Fig. 1, so that the transmitted sound was in the form of a traveling plane wave. Again, the velocity u in the orifice was observed by means of the hot wire, and the "driving" sound pressure p_1 in the cavity, as well as the transmitted sound pressure p_2 , was determined.

Typical oscilloscope traces of these quantities are shown in Fig. 8. At low driving sound pressures, with levels of 120 dB and below, we see that p_1 , u , and p_2 are essentially sinusoidal. The velocity is 90° out of phase with p_1 but in phase with p_2 , as expected. Actually, the measured magnitude of p_2 is found to be consistent with the value $p_2 = (A_o/A)(\rho c)u$, which is the theoretically expected result for p_2 if we assume that the velocity is uniform over the orifice (A_o is the orifice area and A is the duct area).

The situation is illustrated in more detail in Fig. 9, in which the pressure and velocity scales have been chosen in such a manner that u and $p = (A_o/A)(\rho c)u$ coincide. The traces of the actual velocity and the transmitted sound pressure p_2 at a driving sound pressure p_1 corresponding to a level of 160 dB are shown as solid lines. The dashed curve represents the velocity or pressure curve that would have resulted if the orifice impedance had retained its linear value at this level; the curves coincide, since in the linear régime the relation p_2

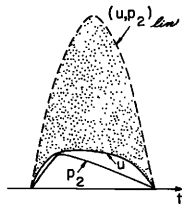


FIG. 9. Dashed curve $(u, p_2)_{lin}$ shows the velocity that would have resulted in the orifice at a driving SPL of 160 dB if the orifice impedance had remained linear at this level. This curve should be compared with the actual velocity curve. Transmitted sound-pressure curves are also shown. The sound-pressure scale is chosen so that in the linear régime the velocity and pressure curves coincide. Note that in the nonlinear régime the same proportionality constant does not apply over the entire cycle.

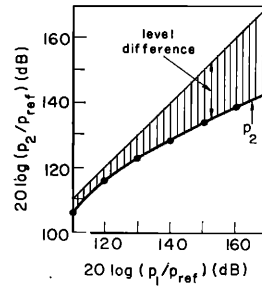


FIG. 10. SPL (p_2) transmitted through an orifice in a plate as a function of the driving sound pressure (p_1). Also shown, by the shaded portion, is the difference in level of p_1 and p_2 .

$= (A_o/A)(\rho c)u$ applies throughout the entire cycle. In the nonlinear region, however, this relation appears to apply approximately only to the peak value of the pressure. During the rest of the cycle, the pressure curve is smaller than $(A_o/A)\rho c$ times the instantaneous velocity. This behavior can be interpreted as a decrease in acoustic-radiation efficiency at higher levels because part of the velocity field in the vicinity of the orifice is no longer irrotational and does not contribute to the sound field.

In this context, it is interesting to study the pressure drop across the orifice, and in Fig. 10 we have shown the measured level of the fundamental-harmonic component of the transmitted sound pressure p_2 as a function of the level of the driving sound pressure p_1 . We have also indicated the level difference between p_1 and p_2 . Although the level difference amounts to only 5 dB at low levels, it reaches values of greater than 20 dB at ~ 160 dB. It should be emphasized that the level difference given in Fig. 10 refers to the fundamental-frequency component. Because of the orifice nonlinearity, however, the transmitted sound pressure also contains overtones. They are predominantly of odd order, as can be seen from the spectrum in Fig. 11, which refers to a SPL of 162 dB of the driving sound pressure.

The measured values of p_1 and p_2 are related to the orifice impedance by the expression

$$\left(\frac{p_1}{p_2}\right)^2 = \left(\frac{R}{\rho c} \frac{A}{A_o} + 1\right)^2 + \left(\frac{X}{\rho c} \frac{A}{A_o}\right)^2. \quad (4)$$

In the high-amplitude régime when orifice reactance is

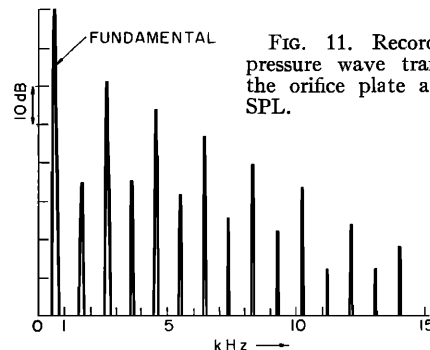


FIG. 11. Recorded spectrum of pressure wave transmitted through the orifice plate at 162-dB driving SPL.

small as compared with the resistance R , we have

$$\frac{R}{\rho c} \sim \frac{A_o}{A} \left[\left(\frac{p_1}{p_2} \right) - 1 \right], \quad R \gg X, \quad (5)$$

where A_o and A are the orifice and duct areas. In the linear region, on the other hand, R is negligible, and we get

$$\frac{X}{\rho c} \sim \frac{A_o}{A} \left[\left(\frac{p_1}{p_2} \right)^2 - 1 \right]^{\frac{1}{2}}, \quad X \gg R. \quad (6)$$

Using the data given in Fig. 10, we have determined from Eqs. 5 and 6 the orifice impedance $|Z|$ as a function of the velocity amplitude in the orifice as given in Fig. 12.

This impedance at low levels is in better agreement with the calculated value of the orifice reactance than the previous results, which, as already mentioned, were not particularly accurate at low levels. At high levels, the impedance is in better agreement with previous results, but is still somewhat smaller. Although some of this discrepancy may be explained by an error in SPL measurement of ± 1 dB, there appears to be a consistent trend that the measurements performed in the presence of the terminating-duct section (see Fig. 1) give somewhat smaller values of the nonlinear resistance than with the tube removed. This is even more pronounced in the presence of a steady superimposed flow. Thus it appears that the "wall effect" produced by the tube termination serves to reduce somewhat the acoustic nonlinear resistance.

An interesting question concerns the frequency dependence of the nonlinear orifice impedance at levels when the resistance is the dominant part of the impedance. In this region, we use the approximate expression $R \approx \rho u_o$ and assume that this result is frequency-independent. Measurements of the frequency dependence of the velocity amplitude in the orifice at a constant driving sound pressure in the high-amplitude region have substantially confirmed this frequency independence of the nonlinear orifice resistance at least up to a frequency at which the orifice reactance equals the resistance.

IV. FREQUENCY-RESPONSE MEASUREMENTS

As we have indicated, the hot-wire method of measuring the acoustic-orifice impedance, discussed in Sec. II, is not particularly suitable for the measurement of small values of the orifice resistance. Therefore, in order to obtain some data on the resistance in the low-level region, we have measured the frequency response of the Helmholtz resonator formed by the orifice plate and the cavity that it covers, with the external tube termination absent (see Fig. 1). The orifice resistance and reactance are then determined from measurements of the bandwidth and the resonance frequency. This method,

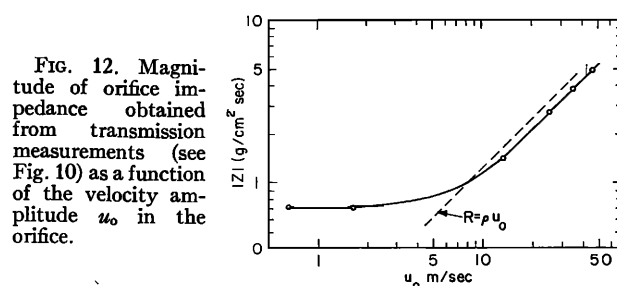


FIG. 12. Magnitude of orifice impedance obtained from transmission measurements (see Fig. 10) as a function of the velocity amplitude u_o in the orifice.

although best suited for small values of the orifice resistance, in contrast to the previous methods, also yields information about orifice impedance at high SPL's, but at these levels the analysis of the data is more involved. The particular resonator response that we have studied referred to the frequency dependence of the velocity amplitude u_p of the driving piston required to keep the pressure amplitude in the cavity constant independent of frequency. If we introduce the frequency parameter $\eta = \omega/\omega_0$ and the damping factor of the resonator $\alpha = R/X_0$, where R is the orifice resistance and X_0 is the reactance at the resonance frequency of the resonator, we can express this frequency dependence of the velocity amplitude as

$$u_p^2 = (p/X_0)^2 [(1 - \eta^2) + \alpha^2 \eta^2] / (\eta^2 + \alpha^2), \quad (7)$$

where $\eta = (\omega/\omega_0)$, $\alpha = R/X_0$; and $X_0 = \omega_0 \rho (t + \delta)$, where t is the orifice neck length and δ = "end correction" $\approx 0.85d$ (d = orifice diameter). The resonator frequency is obtained from the well-known expression,

$$\omega_0 = c[A_o/(t + \delta)V]^{\frac{1}{2}}, \quad (8)$$

where V is the cavity volume, A_o the orifice area, and $(t + \delta)$ the total equivalent neck length of the orifice. Actually, we use this expression to determine $(t + \delta)$ and, hence, X_0 from the experimentally determined resonance frequency ω_0 . At small values of $\alpha = R/X_0$, this resonance frequency ω_0 is the same as the frequency $\omega_{\min}$ at which u_p is a minimum. For an arbitrary value of α , the minimum of u occurs at the frequency

$$\omega_{\min} = \omega_0 [(1 + 2\alpha^2)^{\frac{1}{2}} - \alpha^2]^{\frac{1}{2}}, \quad (9)$$

which, for $\alpha \ll 1$, gives $\omega_{\min} \approx \omega_0$. Under the same conditions, $\alpha \ll 1$, we see from Eq. 7 that, when $\omega_{\pm} = \omega_0 [1 \pm (\alpha/2)]$, the u_p value is $\sqrt{2}$ times larger (3 dB) than the minimum value. Thus, measurement of the bandwidth $(\omega_+ - \omega_-)/\omega_0$ determines α and, hence, the resistance R .

If the condition $\alpha \ll 1$ is not satisfied, the procedure for determining R from the resonance curve is more involved. Actually, it follows from Eqs. 9 and 7 that, if α exceeds the value $(1 + \sqrt{2})^{\frac{1}{2}}$, which means

$$R \geq X_0 (1 + \sqrt{2})^{\frac{1}{2}} \approx 1.55 X_0 \quad (\text{nonresonant behavior}), \quad (10)$$

the velocity u_p no longer has a minimum but increases monotonically with frequency, and the resonant charac-

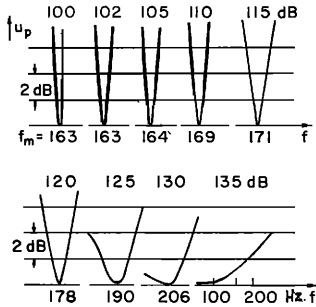


FIG. 13. Frequency dependence of the velocity amplitude u_p of the driving piston (see Fig. 1) required to keep the cavity SPL constant at 110 dB. f_m is the frequency at the minimum point of u_p .

ter of the curve is lost. It should be pointed out that the variation with the velocity amplitude of the location $\omega_{\min}$ of the minimum of the response curve depends not only on the possible dependence of ω_0 on the velocity amplitude. As can be seen from Eq. 9, there is a competing change caused by the increase in the damping α with amplitude that becomes important as α approaches unity. For $\alpha \ll 1$, we have $\omega_{\min} \approx \omega_0 [1 - (\alpha^2/4)]$ and the variation of $\omega_{\min}$ with amplitude is then largely determined by the corresponding variation in ω_0 .

In Fig. 13 is shown a set of measured frequency curves for the piston velocity for some different values of the sound pressure p_1 in the cavity corresponding to the conditions referred to in connection with Eq. 7. The measured resonance frequency at low levels is found to be 163 Hz. With a cavity volume 640 cm³ and an orifice diameter of 0.7 cm, we obtain from Eq. 8 the total equivalent neck length $(l + \delta) \approx 0.68$ cm and $X_0 = 0.86$ g/cm² sec. Using those response curves in Fig. 13 that exhibit pronounced resonant behavior, we find that the measured values of $\alpha = (\omega_+ - \omega_-)/\omega_0$ range from 0.08 to 0.5 in the sound-pressure range from 100 to 125 dB. The corresponding values of the resistance are given in Fig. 14. Similarly, the velocity dependence of the orifice-mass end correction in Eq. 8 is determined from the measured velocity dependence of ω_0 . The corresponding velocity dependence of the reactance is also shown in Fig. 14.

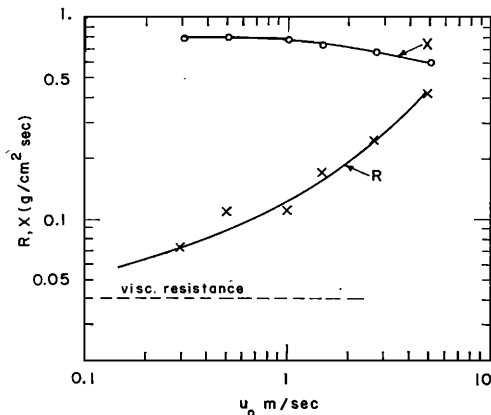


FIG. 14. Orifice reactance X and resistance R as a function of the velocity amplitude u_0 in the orifice. Results are obtained from the frequency-response measurements in Fig. 13.

It is interesting also to compare the nonresonant response curve in Fig. 13 with the condition given in Eq. 10 for nonresonant behavior. Thus, if we use $X_0 = 0.86$ g/cm² sec given above and $R \approx \rho u_0$, which, as we have seen, is approximately valid in the nonlinear region, the condition in Eq. 10 becomes $(u/c) \geq 0.032$, or $u \geq 10.8$ m/sec. From the sound pressure/velocity relation in Fig. 3, we see that this corresponds to $p \geq 135$ dB. In other words, the theoretical condition in Eq. 10 for nonresonant behavior of the resonator response corresponds to a SPL in the cavity of 135 dB. This is in excellent agreement with the experimental result in Fig. 13, where the nonresonant behavior of the response curve is seen to start at a SPL close to the predicted value of 135 dB.

Another set of response curves was also determined in which the piston velocity was held constant independent of frequency and the frequency dependence of the SPL in the cavity was recorded. Again, curves similar to those in Fig. 13 were obtained.

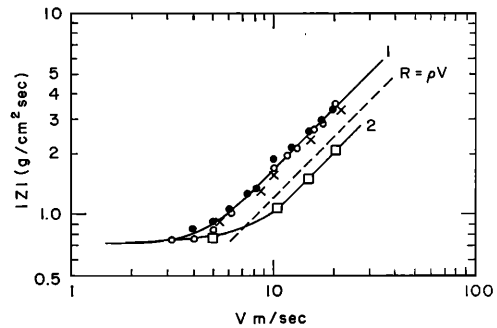


FIG. 15. Influence of steady flow through the orifice on the magnitude of the orifice impedance. Curve 1 refers to measurements in the absence of the external duct. Curve 2 is obtained with the external duct present. Measurements are made with the cavity SPL constant. For the data in Curve 1, three different SPL's were used: 115 dB (●), 120 dB (×), and 125 dB (○). Curve 2 corresponds to a SPL of 115 dB.

V. EFFECT OF SUPERIMPOSED STEADY FLOW

As already indicated in Fig. 1, a superimposed steady flow can be produced through the orifice by suction through an outlet in the cavity wall in front of the orifice. We consider first the case in which the steady-flow velocity is larger than the oscillatory-velocity amplitude and the level of the driving sound pressure is small enough so that, in the absence of steady flow, the relation between sound pressure and velocity is linear. The oscillatory-flow velocity in the orifice is then essentially sinusoidal. As in the absence of steady flow, we made two sets of hot-wire measurements. In one set with the external tube removed, we measured the oscillatory-velocity amplitude in the orifice as a function of the steady-flow velocity at a constant SPL in the cavity. Actually, measurements were made at three different SPL's—115, 120, and 125 dB—and the cor-

responding values of the magnitude of the impedance are shown in Fig. 15.

The second set of measurements was made with the external-tube section attached to the orifice plate. At a driving SPL of 110 dB, the velocity in the aperture, as well as the transmitted sound-pressure amplitude, was measured at different steady-flow velocities up to 20 m/sec. As in the analogous experiment in the absence of the steady flow, the transmitted sound-pressure amplitude and the velocity amplitude in the orifice were found to be consistent with $p_2 = (A/A_0)(\rho c)u$, within experimental accuracy. In other words, the presence of the vorticity in the jet flow produced by the steady flow does not seem to influence markedly the sound emitted by the superimposed acoustic oscillations. The measured difference between the SPL of 110 dB in the cavity and the level of the transmitted sound pressure p_2 is shown as a function of the flow velocity in Fig. 16. On the basis of these results, we have determined the orifice impedance from Eqs. 5 and 6, and the corresponding

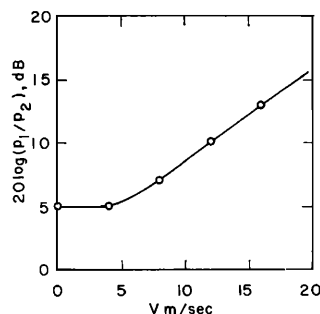


FIG. 16. Drop in SPL across an orifice plate as a function of the steady-flow velocity through the orifice.

magnitude of the orifice impedance $|Z|$ is shown as a function of the steady-flow velocity in Fig. 15. From the two curves in this Figure, we note that the resistance values at high velocities, obtained with the external duct attached to the orifice plate, are smaller than with the duct absent. This effect is more pronounced than, but consistent with, the corresponding measurements of the nonlinear resistance in the absence of flow. At high velocities, when the reactance of the orifice plays a minor rôle, the orifice impedance (resistance) is proportional to the steady-flow velocity. For comparison, we have shown in Fig. 15 (dashed line) also the resistance line $R = \rho V$. We discuss the quantitative results in more detail in Sec. VI.

These results all refer to the case in which the oscillatory velocity u is small as compared with the steady velocity V . In this case, u is practically sinusoidal. If, on the other hand, we consider the case in which u is of the same order of magnitude as V or less, the velocity u is considerably distorted, particularly when the driving sound pressure is large. To illustrate this, we have shown in Fig. 17 oscilloscope traces of u at a driving SPL of 150 dB at some different values of V smaller than u . The dashed base line in this Figure indicates the zero value of the total velocity, and the dashed portion

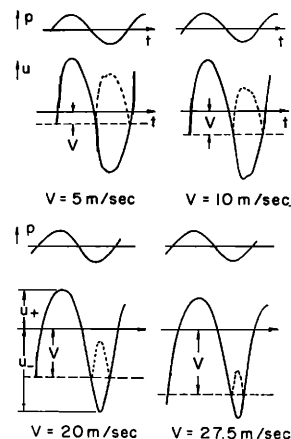


FIG. 17. Oscilloscope traces of the sound pressure in the cavity and the velocity in the orifice at some different values of the superimposed steady-flow velocity V . The SPL in all cases is 150 dB.

of the curve is drawn to indicate the rectified hot-wire trace as it appeared on the oscilloscope. We have reflected this portion in the base line to obtain the actual velocity curve. It is interesting to note the behavior of the velocity curve when the steady-flow velocity is about equal to the oscillatory velocity amplitude. Then, in the part of the cycle in which u and V are in the same direction, the oscillatory amplitude is considerably reduced below its linear value, since the total velocity is then inside the nonlinear region. During the other part of the cycle, on the other hand, the total velocity is close to zero and the sound pressure/velocity relation is about the same as in the linear region. Therefore, the negative excursion of the velocity curve is considerably larger than the positive; this results in a distortion of the oscillatory curve. Because of this behavior, an increase of the steady-flow velocity V in this region does not lead to a noticeable reduction of the peak-to-peak value of the oscillatory velocity and the amplitude of the fundamental harmonic component (which is half the peak-to-peak value).

The influence of a superimposed flow on the resonator characteristics given in Eq. 7 was also studied. Some of the response curves for the different values of the steady-flow velocity at a 110-dB SPL level are shown in Fig. 18. As expected, the steady flow increases the damping, and at a sufficiently large value of the velocity the orifice resistance is large enough to eliminate the resonant character of the response curve, in accordance with the condition in Eq. 10. An analysis of these response curves in the same manner as described

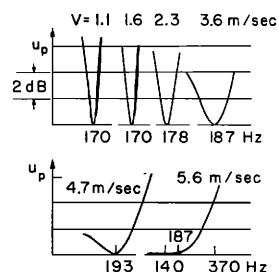


FIG. 18. Frequency curves of the velocity amplitude u_p of the driving piston (see Fig. 1) required to keep the SPL in the cavity constant at 110 dB at different dc flow velocities V .

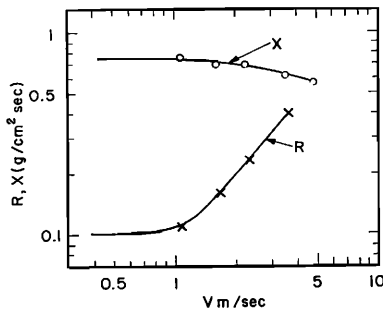


FIG. 19. Influence of steady-flow velocity on the orifice reactance and resistance as obtained from frequency-response curves illustrated in Fig. 18.

previously yields the values of orifice resistance and reactance indicated in Fig. 19 as a function of the steady-flow velocity. As before, the reactance refers to a frequency of 150 Hz. Since these measurements were made at 110 dB SPL, the resistance for values of V less than 1 m/sec should approach the resistance value 0.1 that corresponds to 110 dB in the absence of flow; we have extrapolated the results in Fig. 19 accordingly.

VI. ORIFICE IMPEDANCE—SUMMARY AND DISCUSSION

We have presented data at low and high SPL's on the nonlinear behavior of the orifice impedance and on the influence on superimposed steady flow through the aperture. We now combine these results to obtain the behavior of the orifice impedance over an extended range of velocities. Considering first the nonlinear acoustic impedance in the absence of steady flow, we summarize in Fig. 20 the data in Figs. 7 and 14 that refer to the case in which there is no tube section attached to the outside of the orifice plate. Thus, in Fig. 20, the dotted lines refer to the hot-wire method of measurement, whereas the dashed lines correspond to

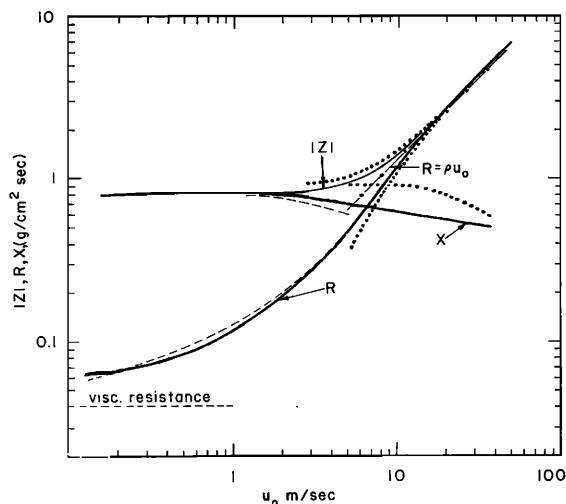


FIG. 20. Summary of the data on the nonlinear behavior of acoustic impedance of an orifice from frequency-response measurements at low SPL's (---) and hot-wire measurements at high SPL's (...). For comparison, the line $R = \rho u_0$ is also shown. u_0 : Velocity amplitude in orifice.

the measurements of resonator characteristics. On the basis of these curves, we have constructed the "average" curves (solid lines) for $|Z|$, R , and X . Although the resistance data from the two methods of measurements match up quite well, there is a marked discrepancy in the reactance curve. This discrepancy is likely to be an error at low levels in the hot-wire impedance curve, which approaches a value that is approximately 20% higher than the theoretically predicted and the value obtained from the resonator measurements. Therefore, in drawing the average-reactance curve, we have given the resonator data more weight.

Before discussing the properties of this curve, it is interesting to compare it with previous results based on still another method of measurement. Thus, in Fig. 21,

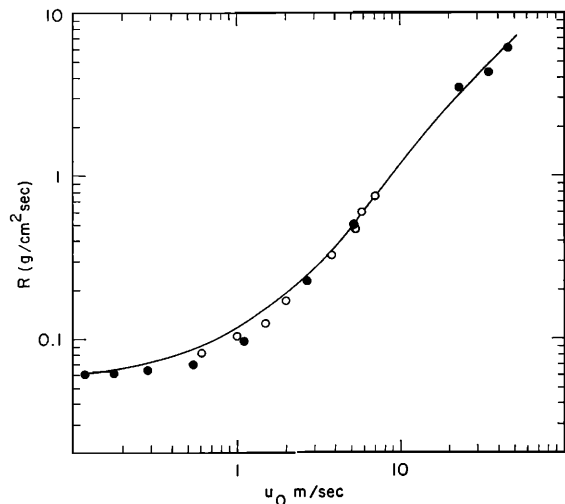


FIG. 21. Comparison of present orifice-resistance data with previously reported results based on impedance-tube measurements of orifices with different dimensions. In the present study (—), the orifice diameter $d = 0.7$ cm and plate thickness $t = 0.01$ cm at the orifice. The data of Bies and Wilson (●; Ref. 4) refer to $d = 9$ cm, $t = 2.6$ cm. The data of Ingard (○; Ref. 3) refer to $d = 1.4$ cm, $t = 0.32$ cm.

we have shown results obtained from the impedance-tube measurements by Bies and Wilson⁴ and by Ingard.³ In terms of the notation for orifice resistance R and orifice area A_0 used here, Bies and Wilson presented (R/A_0) in their paper as a function of the root-mean-square value of the velocity in the orifice. Their data refer to an orifice plate mounted as part of a Helmholtz resonator terminating a tube with the plate perpendicular to the tube axis. The orifice diameter was 3.5 in., and the thickness of the orifice plate was 1.3 in. The frequency was in the range 60–160 Hz. Thus, multiplying the Bies-Wilson resistance data by A_0 and accounting for the difference in root-mean-square and peak values of the velocity, we obtain the values indicated in Fig. 21. Ingard's measurements were limited to the velocity amplitudes in the orifice less than 10^3 cm/sec. He presented his resistance data in terms

of a nonlinear resistive "end correction," Δ_{NL} , and plotted Δ_{NL}/d as a function of the velocity amplitude in the orifice. This quantity is defined in terms of the total orifice resistance R by the relation³

$$R \simeq (8\mu\omega)^{1/2}(1+t/d+\Delta_{NL}/d), \quad (11)$$

where t is the thickness of the orifice plate, d is the orifice diameter, μ is the viscosity coefficient, ρ is the air density, and $\omega/2\pi$ is the frequency. The corresponding data points obtained from the reported values of Δ_{NL}/d for an orifice plate with $d=1.4$ and $t=0.32$ cm are also shown in Fig. 21.

Apart from a small discrepancy at low levels, we see that, on the whole, there is good agreement between the present impedance curve and the results of Bies-Wilson and Ingard. This agreement is particularly striking, since the orifices involved are widely different, with orifice diameters ranging from 0.7 to 9 cm.

At low levels, where viscous drag dominates ($\Delta_{NL}=0$) over the orifice resistance, we see from Eq. 11 that, if the plate thickness t is small as compared with the orifice diameter d , the linear orifice resistance is nearly independent of quantities t , d . This condition is approximately satisfied for the orifice plates involved here, and, apart from differences in the frequencies at which the experiments were carried out, the consistency of the data in Fig. 21 at low SPL's for the different orifices can be understood on the basis of Eq. 11.

Also, at high levels, the independence of the resistance on the orifice dimensions can be understood if we accept the result that in this region the sound pressure/velocity relation is given by $p \simeq \rho u_o^2$, with a corresponding orifice resistance given by

$$R \approx \rho u_o, \quad u_o \gg u_c = (1+t/d)(8\omega\mu/\rho)^{1/2}. \quad (12)$$

This result can be understood on the basis of Bernoulli's law if we assume that the sound pressure in the orifice is a certain fraction q of p_1 in the cavity. We then have $(1-q)p_1 = (\rho u_o^2/2)(1-\sigma^2)$, where $\sigma = A_o/A$ is the ratio between the orifice area and the duct area. If we set $q \approx \frac{1}{2}$, which is quite reasonable since we expect the sound pressure in the orifice to be approximately the average of the sound pressures on the two sides of the aperture,¹² we get $p_1 \simeq \rho u_o^2(1-\sigma^2)$ and $R = p_1/u_o \simeq \rho u_o(1-\sigma^2)$, in agreement with our result, since in our case $\sigma \ll 1$. However, if σ^2 cannot be neglected, this argument suggests that, to make Eq. 12 more general, we should multiply the expression for R by $(1-\sigma^2)$. Equation 12 is valid in the velocity region in which the viscous contribution to the resistance given by Eq. 11

is small, that is, when $\rho u_o \gg \rho u_c$, where

$$u_c = (1+t/d)(8\omega\mu/\rho)^{1/2}. \quad (13)$$

Experimentally, we find that the condition $u_o \gg u_c$ is satisfied when $u_o \geq 25u_c$ or $u_o \geq 25(8\omega\mu/\rho)^{1/2}[1+(t/d)] = 10^3[(f/200)^{1/2}(1+(t/d))]$, which at a frequency $f=200$ Hz is 10^3 cm/sec, if we use the kinematic viscosity for air, $\mu/\rho=0.15$.

In regard to the orifice reactance, for which less data are available about the nonlinear behavior, the linear value is of the form³

$$X = \omega\rho(t+\delta), \quad u < u_c, \quad (14)$$

where δ , the mass end correction of the orifice, is approximately equal to $\delta \simeq 0.85d$ when $\sigma = A_o/A$ is much less than unity. When σ is not small, the expression for δ is more complicated³ and goes to zero when $\sigma=1$. At high velocities, $u \gg u_c$, which according to our experimental results means $u_o > 10$ m/sec, the mass reactance decreases and approaches a value that we expect to be approximately one-half the linear value. This expectation is based on the assumption that, in the high-velocity region, the flow at any moment is irrotational only on the side of the orifice where at that moment the flow is toward the orifice. Thus, associating the acoustic inertia of the flow with the irrotational portion leads to the reduction factor of approximately one-half.

On the basis of these results regarding the nonlinear behavior of the orifice impedance, we have seen that a unified description, only weakly dependent on the orifice geometry, can be obtained if we use as orifice impedance the ratio between sound pressure and velocity in the orifice (and not this ratio divided by the orifice area as used by Bies and Wilson), and we propose that the curves in Fig. 20 represent this unified description. At low levels, $u_o < u_c$, the resistance is given by Eq. 11 and the reactance by Eq. 14. At high-orifice velocity amplitudes, $u_o \geq (f/200)^{1/2}[1+(t/d)] 10^3$ cm/sec, where f is the frequency, the resistance is given by Eq. 12. In the transition region, the resistance approaches the high-level asymptotic value in Eq. 12 from below.

With regard to the influence of the steady flow on the orifice impedance, we have summarized in Fig. 22 the orifice-impedance data expressing the influence of steady flow on the orifice impedance. We have used the data in Figs. 15 and 19 referring to the experiments with no external-tube section attached to the orifice plates. The dotted and dashed lines correspond to the hot-wire and resonator experiments, respectively. In the hot-wire experiments, only values of $|Z|$ were obtained. The linear value of the reactance is consistent with the theoretical value, and decreases with increasing velocity. The resonator data on the resistance were taken at 110 dB SPL. As the steady-flow velocity V approaches the value of the oscillatory-velocity amplitude, which at 110 dB SPL in the cavity is ~ 0.8 m/sec, the re-

¹² It should be pointed out in this context that the acoustic-pressure distribution generally cannot be assumed to be the same as the pressure distribution obtained for steady jet flow through the orifice. The characteristic time required to establish steady-state pressure distribution involves the characteristic time of diffusion of vorticity L^2/ν , where L is the characteristic jet or cavity dimension and ν is the kinematic viscosity. This time is much longer than the acoustic periods involved here.

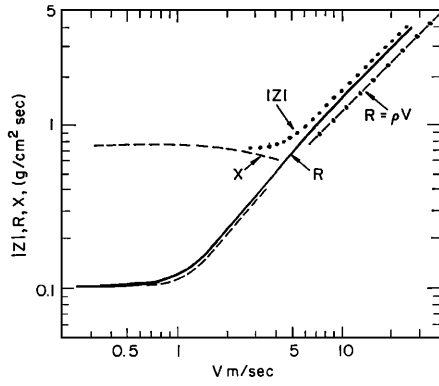


FIG. 22. Summary of the data on the influence of steady-flow velocity V on the orifice impedance from frequency-response measurements (---) and from hot-wire measurements (...). For comparison, the curve $R = \rho V$ is shown. SPL = 110 dB.

sistance must approach the resistance value at 110 dB in the absence of steady flow. This resistance, according to the data in Fig. 20, is ~ 0.11 . We have extrapolated the resistance curve in Fig. 22 accordingly.

It is interesting to note that the influence of the steady flow on the orifice impedance at high levels is quite similar to the influence of the velocity amplitude u_0 on the nonlinear impedance in the absence of steady flow. Thus, at sufficiently high velocities, the resistance is proportional to V so that $R \approx C\rho V$, where, according to our measurements, the constant C is approximately unity when the external-tube section is present and somewhat larger when the tube is absent. Actually, if we let the resistance for $V=0$ be R_0 , it is a good approximation to set

$$R \approx C\rho V, \quad V \geq R_0/\rho, \quad (15)$$

for velocities larger than R_0/ρ , where C lies between 1 and 1.5. Again, we can interpret this result from Bernoulli's law. Thus if we let the sound pressure in the orifice (where V is measured) be a fraction q_1 of the driving sound pressure p_i , we obtain¹³ $R \approx \rho V(1-\sigma^2)/(1-q_1)$. Thus, to get agreement with our result, we need to set $q_1 = \frac{1}{3}$, which gives $R \approx 1.5 \rho V$, since $\sigma = A_o/A \ll 1$ in our case.

VII. APPLICATION. RESONATOR ABSORBER FOR HIGH-INTENSITY SOUND

In the light of the results obtained regarding the acoustic nonlinearity of an orifice, it is interesting to consider some aspects of the response of an ordinary Helmholtz resonator exposed to an incident high-intensity sound wave. In particular, we consider a resonator that forms a termination of a duct. The analysis of this case applies equally well to the important practical problem dealing with the response to

¹³ When the "percentage open area" $\sigma = A_o/A$ of the orifice plate is not negligible, this argument suggests that the expression for the resistance in Eq. 15 should be multiplied by the factor $(1-\sigma^2)$.

an incident-plane sound wave of a resonator type of absorptive treatment on a wall. Usually such a wall treatment consists of a perforated thin plate spaced a certain distance from a rigid wall by means of a cell structure, as indicated schematically in Fig. 23.

The boundary impedance of the resonator termination or the wall treatment—that is, the ratio of the plane-wave pressure in the duct and the corresponding average velocity in the plane of the resonator plate—can be expressed as $Z_b = (Z/\sigma) + Z_o$, where σ is the ratio between the orifice area and the duct area, Z is the orifice impedance previously discussed, and Z_o is the impedance of the air cavity behind the orifice plate. The resistive part of Z_b equals (R/σ) , where R is the orifice resistance. If we use the experimentally found value for R as given in Eq. 12, modified by the factor $(1-\sigma^2)$ as discussed in Sec. VI, we have

$$(Z_b/\rho c) = (u_o/c\sigma_1) - i\chi = \theta - i\chi, \quad (16)$$

$$\sigma_1 = \sigma/(1-\sigma^2)$$

where χ is the sum of the orifice reactance and the impedance of the air cavity behind the plate. The absorption coefficient that corresponds to this boundary impedance is

$$\alpha = 4(u_o/c\sigma_1)/[(u_o/c\sigma_1) + 1]^2 + \chi^2. \quad (17)$$

But the velocity amplitude u_o in the orifice is not known *a priori*, and, in order to obtain the dependence of the absorption coefficient on the sound-pressure amplitude p_i of the incident wave and on the frequency, we must determine u_o in terms of p_i from the relation $\sigma u_o = 2p_i/(Z_b + \rho c)$ or

$$(\sigma u_o)^2 = 4\left(\frac{p_i}{\rho c}\right)^2 \left[\left(\frac{u_o}{c\sigma_1} + 1 \right)^2 + \chi^2 \right], \quad (18)$$

which follows directly from the boundary conditions. For simplicity, we consider the absorption curve only in the neighborhood of the resonance frequency. At the resonance frequency, we have $\chi=0$ and the relation between u_o and p_i is then

$$(u_o/\sigma_1 c)_{\text{res}} = \left[\frac{1}{4} + 2(p_i/\gamma P_0 \sigma_1) \right]^{\frac{1}{2}} - \frac{1}{2}. \quad (19)$$

We have introduced $c^2 = \gamma P_0/\rho$, where P_0 is the static pressure, and $\gamma = c_p/c_v$. Using this expression for $(u_o/c\sigma_1)$ in the denominator in Eq. 17, we obtain a first

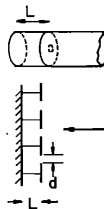


FIG. 23. Nonlinear impedance data applied to study the absorption of high-intensity sound by a resonator array.

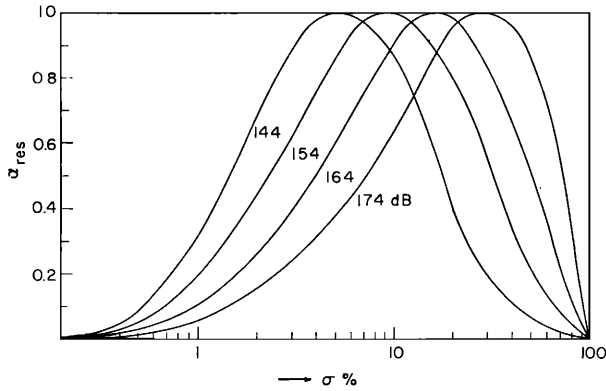


FIG. 24. Absorption coefficient at resonance for high-intensity sound incident on a resonator wall treatment (see Fig. 23) as a function of the percentage open area $\sigma = A_0/A$ of the resonator. The parameter is the SPL in decibels of the incident wave.

approximation of the frequency dependence of u_0 in terms of p_i

$$u_0/\sigma_1 c = 4\beta / \{ [(1+8\beta)^{1/2} + 1]^2 + 4\chi^2 \}^{1/2} \quad (20)$$

where $\beta = p_i/\gamma P_0 \sigma_1 = p_i(1-\sigma^2)/\gamma P_0 \sigma^2$. To obtain the corresponding frequency dependence of the absorption coefficient, we first note that from Eqs. 17 and 18 follows $\alpha = (u_0/c\sigma_1)(\sigma u_0)^2/(p_i/\rho c)^2$. Then we use Eq. 20 to obtain

$$\alpha \approx 64\beta / \{ 1 + (1+8\beta)^{1/2} \}^2 + 4\chi^2 \}^{1/2}, \quad (21)$$

where $\beta = (p_i/\gamma P_0)[(1-\sigma^2)/\sigma^2]$. The absorption at resonance,

$$\alpha_{\text{res}} = 64\beta / [1 + (1+8\beta)^{1/2}]^3, \quad (22)$$

varies comparatively slowly with the parameter β . It remains above 0.8 when β varies from 0.3 to 5, reaching 1 when $\beta=1$. For example, to get 100% absorption at resonance of an incident SPL of 160 dB, corresponding to a pressure amplitude of $2.8 \cdot 10^4$ dyn/cm², the resonator absorber should have a percentage open area $\sigma \approx (p_i/\gamma P_0)^{1/2} \approx 14\%$. The resonance absorption coefficient for other values of σ is plotted in Fig. 24 for some different levels of the incident sound wave.

To study the frequency response of α as given by Eq. 21, we need the explicit form of the reactance χ . If the depth of the air cavity behind the orifice plate is L , the expression for χ —if L is small as compared with the wavelength—is

$$\chi = (k_0 L)^{-1} [(\omega/\omega_0) - (\omega_0/\omega)], \quad L/\lambda \ll 1. \quad (23)$$

The resonance frequency ω_0 is given by $\omega_0 \rho(t+\delta) = (1/k_0 L)$, where $(t+\delta)$ is the effective thickness of the orifice plate and $k_0 = \omega_0/c$. If $L \ll \lambda$ is not fulfilled, we must use the general expression

$$\chi = (\omega/\omega_0) \cot k_0 L - \cot [(\omega/\omega_0) k_0 L]. \quad (24)$$

Restricting ourselves to the case given in Eq. 23, we can express the bandwidth of the absorption curve by

$$\Delta\omega/\omega_0 = (k_0 L/2) [1 + (1+8\beta)^{1/2}],$$

where $\Delta\omega$ is the width of the absorption curve, with $\alpha = \alpha_{\text{max}}/(\sqrt{2})^3$.

It should be emphasized that the resonator absorption characteristics discussed here apply only to high SPL's that in our experiment correspond to velocity amplitudes in the orifice larger than 10 m/sec. Knowing this lower bound of the velocity amplitude, we can use Eq. 19 to obtain the corresponding lower bound of the pressure amplitude. For example, with an orifice diameter of 0.5 cm and a percentage open area $\sigma = 0.05$, we find a SPL lower bound of 140 dB.

ACKNOWLEDGMENTS

This work was begun when one of us (U.I.) was a visitor at the Institut für Technische Akustik at the Technische Universität of Berlin, during the summer of 1966, and he wishes to thank the director of the Institut, Professor Lothar Cremer, and his staff, particularly M. Hubert and P. Rother, for the hospitality that he enjoyed during his stay. Many thanks go also to D. Bechert, of the Institut für Turbulenzforschung, for his kind help in connection with hot-wire instrumentation.

This work was supported in part by the U. S. Navy (Office of Naval Research) under contract.